

Enterprise Microclimate Heat Intelligence Platform (V1 Stable & V2 Thermora)

VERSION 1 (STABLE)

VERSION 2 (THERMORA)

GOOGLE GENAI ENGINE

FORTYGUARD IOT FLEET

ISO 7243 WBGT STANDARD

SECTION 1: Executive Overview & Public Production Deployments

HEATSHIELD AI is an authoritative, multi-agent urban heat resilience platform designed to safeguard municipal workforces, transit corridors, and high-density urban sectors from extreme heat events. The platform directly addresses the limitations of macro-level weather forecasts by ingesting high-resolution microclimate telemetry from the **FortyGuard Temperature API**, calculating physiological heat stress indices (ISO 7243 WBGT, NOAA Heat Index, UTCI), and driving automated multi-agent operational mitigation workflows.

Verified Public Production Deployments

Both platform versions are concurrently hosted and accessible in real-time:

Platform Version	Engine & Architecture Description	Verified Public URL
Version 1 (Stable)	Baseline Heat Analytics Engine & Tactical Command Center with deterministic emergency dispatch.	https://heatshield-0a6n.onrender.com/?v=1
Version 2 (Thermora)	Dynamic AI Heat Intelligence Engine & Multi-Agent Studio featuring stateful multi-turn Gemini reasoning.	https://heatshield-0a6n.onrender.com/?v=2

Synchronized Version Switching: The top navigation bar and bottom chat model selector pill feature two-way synchronization. Toggling between *V1 Stable* and *V2 Thermora* updates UI themes, prompt suggestions, and the chatbox input placeholder in real-time.

SECTION 2: Dual-Version Architecture & Feature Comparison

🔥 Version 1 (Stable) — Baseline Tactical Hub

- Core Purpose:** Real-time situational awareness, deterministic emergency dispatch, and baseline monitoring.
- UI Styling:** Clean light mode card architecture (`bg-white` , `border-slate-300`).
- 24-Hour Diurnal Curves:** High-resolution telemetry plot (Asphalt Surface: 48.2°C, Air Temp: 38.6°C, WBGT: 31.4°C).
- Chat Input Placeholder:** "Ask V1 Stable about thermal stress, cooling simulations, or OSHA limits..."
- Model Attribution:** Baseline Heat Analytics Engine.

⚡ Version 2 (Thermora) — AI Intelligence Engine

- Core Purpose:** Dynamic AI-driven heat intelligence, OSHA compliance calculation, and predictive physics ML.
- UI Styling:** High-contrast light theme with offset background (`bg-slate-100/80` container wrapper) for sharp visual visibility.
- Conversational Agent:** Multi-turn stateful chat memory (`history: [{ role, parts }]`).
- Key Triggers:** `@` mention menu (5 agents/context) and `/` quick operational commands.
- Chat Input Placeholder:** "Ask V2 Thermora about thermal stress, cooling simulations, or OSHA limits..."

Comprehensive Architectural Matrix

Capability / Dimension	Version 1 (Stable)	Version 2 (Thermora)
Primary AI Model	Deterministic Multi-Agent Dispatcher	Google Gemini 2.5 Flash / Dynamic Orchestrator
Conversation Memory	Single-Turn Direct Telemetry Query	Full Multi-Turn Thread History (Stateful Context)
Thinking Budget Mode	Zero Latency Direct Dispatch	<code>thinkingBudget: 0</code> (No reasoning loops)
Chat UI Theme	Light Border (<code>border-slate-300</code> , <code>bg-white</code>)	High-Contrast Light (<code>bg-slate-100/80</code> wrapper, <code>shadow-md</code>)
Command Popups	Pre-built Prompt Pills	<code>@</code> Agent Mentions + <code>/</code> Quick Action Menu

SECTION 3: UI Guardrails & Design Constraints

To ensure rock-solid production stability and eliminate cognitive clutter, HEATSHIELD enforces the following strict frontend UI constraints:

🔒 1. Strict 2-Option Model Dropdown

The model selector dropdown menu on the chat toolbar has been strictly restricted to list **ONLY TWO** platform engines. Extraneous options (e.g. Gemini 3.7 Flash High, Gemini 2.5 Pro, FortyGuard Neural-v2) have been removed:

1. **V1 Stable** — *Baseline Heat Analytics Engine*
2. **V2 Thermora** — *Dynamic AI Heat Intelligence Engine*

🎤 2. Microphone & Audio Icon Elimination

The microphone/voice recording icon () has been completely removed across both version interfaces to maintain a focused, command-line text interface.

🌟 3. Asterisk & Star Sanitization Protocol

To prevent messy markdown bolding and asterisks in technical reports, HEATSHIELD enforces a dual-layer sanitization protocol:

- **Backend System Instruction:** *"Do not use markdown bolding, asterisks (*), or star decorators anywhere in your output. Return clean, unformatted plain text with plain numbers and line breaks."*
- **Frontend Parser Sanitizer:** All incoming messages pass through `cleanedText = rawText.replace(/\*/g, '')` before DOM insertion.

SECTION 4: API Specifications & Data Contracts

1. Telemetry Pipeline Endpoint (`GET /api/v1/fortyguard/telemetry`)

Fetches real-time sensor node readings across high-risk municipal sectors:

```
// Response Payload Contract
{
  "sector_id": "Sector 7 - Downtown Core",
  "surface_asphalt_temp": 48.2,
  "ambient_air_temp": 38.6,
  "relative_humidity": 58.0,
  "wbgt_stress_index": 31.4,
  "heat_index": 46.8,
  "uv_index": 11.0,
  "risk_level": "EXTREME",
  "timestamp": "2026-08-29T14:30:00Z"
}
```

2. Stateful Multi-Turn Agent Chat (`POST /api/v1/agents/chat`)

Processes natural language inquiries with complete conversation thread history and zero thinking loops:

```
// Request Payload Contract
{
  "user_message": "Calculate OSHA work-rest schedule for Sector 7 outdoor workers",
  "active_zone": "Sector 7 - Downtown Core",
}
```

```
"model_name": "gemini-2.5-flash",
"temperature": 0.7,
"history": [
  { "role": "user", "text": "What is the current asphalt temperature in Sector 7?" },
  { "role": "assistant", "text": "Current asphalt temperature is 48.2 C with WBGT at 31.4 C." }
]
}

// Backend Execution Parameters:
// thinkingConfig: { "thinkingBudget": 0 } -> Reasoning output disabled
```

SECTION 5: Master Antigravity Generation Prompts

The following executable prompts configure automated engineering workflows for repository documentation and API integration:

Prompt 1: Project Summary & Architecture Generator (`PROJECT_SUMMARY.md`)

```
Generate a comprehensive, production-grade PROJECT_SUMMARY.md for the HEATSHIELD AI codebase:
1. Include platform overview, FortyGuard API integration pipeline, and 4 specialized agent roles:
  - Heat Sentinel (WBGT & surface anomaly tracking)
  - Vulnerable Population Advisor (Demographic GIS raster & shelter dispatch)
  - Urban Cooling Strategist (Cool roofs, misting cannons & albedo ML)
  - Grid Balancer (Substation load shaving & thermal pre-cooling)
2. Detail the Dual-Version UI (V1 Stable vs V2 Thermora) with live deployment URLs:
  - V1 Stable: https://heatshield-0a6n.onrender.com/?v=1
  - V2 Thermora: https://heatshield-0a6n.onrender.com/?v=2
3. Document core backend formulas: Stull Wet-Bulb, NOAA Heat Index, and Inverse Distance Weighting (IDW).
4. Output strictly valid markdown with structured tables and architecture diagrams.
```

Prompt 2: API Documentation & Integration Specification (`API_DOCUMENTATION.md`)

```
Generate an authoritative API_DOCUMENTATION.md for all HEATSHIELD FastAPI endpoints:
1. Endpoints to document:
  - GET /api/v1/health -> Health status & uptime
  - GET /api/v1/fortyguard/telemetry -> IoT sensor stream & UHI index
  - GET /api/v1/fortyguard/diurnal -> 24-hour diurnal thermal curves
  - POST /api/v1/heat-intelligence/calculate -> ISO 7243 WBGT & Heat Index
  - POST /api/v1/ml/simulate-mitigation -> Physics-guided cooling reduction
  - POST /api/v1/agents/chat -> Multi-turn stateful conversational agent
  - POST /api/v1/agents/orchestrate -> 4-agent autonomous tactical cycle
2. For each endpoint include: Method, Path, Auth, Query/Body Parameters, Response JSON, and Curl Example.
3. Explicitly state the thinkingBudget: 0 configuration and asterisk sanitization rules.
```